National University of Computer and Emerging Sciences

Technical & Business Writing (SS2007)

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Course Instructor(s)

Atifa Batool, Javeria Ali, Khadija Shereen, Madiha Rehman, & Sameera Sultan

Sessional-I Exam

Total Time: 1 Hour
Total Marks: 35
Total Questions: 02

Semester: SP-2024
Campus: Karachi
Dept: Computer Science

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CLO #3: Design, document, and develop a research project.

Q1: Read the following abstracts and answer the questions that follow. [10 marks]

Abstract 1: The use of emotion to drive robotic interaction continues to grow across a range of use cases, from social robotics to increased survivability. Nevertheless, these efforts remain isolated from each other and are not easily compared between papers and projects. To this end an extensive survey of 1427 IEEE and ACM publications was conducted, covering robotics and emotion. The survey first resulted in broad categorizations of key trends covering emotional input and output. This was followed by an extended analysis on 232 papers that focused on the internal processing of emotion, where emotion was handled through some kind of algorithm and not just as an input or output. From this analysis, three broad categories were developed: emotional intelligence, emotional model, and implementation. Emotional intelligence captured the manner in which emotion was handled and included the subcategories: algorithm, mapping, and history. The emotional model category captured the emotion categories and number of emotions used, while the implementation category tracked the role, purpose, and platform. This paper concludes with a summary of key features discovered through the process, future opportunities, and a discussion of the intrinsic challenges emerging from the interaction of emotion and robotics.

Abstract 2: The majority of Android smartphone apps are free. When an application is used, advertisements are displayed in order to generate revenue. Adware-related advertising fraud costs billions of dollars each year. Adware is a form of advertising-supported software, that turns into malware when it automatically installs additional malware and adware on an infected device, steals user data, and exposes other vulnerabilities. Better techniques for detecting adware are needed due to the evolution of increasingly sophisticated evasive malware, particularly adware. Even though significant work has been done in the area of malware detection, the adware family has received very little attention. This paper presents a deep learning-based scheme called

MadDroid to detect malicious Android adware based on static features. Moreover, this paper delivers a novel dataset that consists of malicious Adware and benign applications and an optimized Convolutional neural network (CNN) for detecting Adware infected by malware based on the permissions of the applications. The results indicate an average classification rate that is higher than previous work for individual adware family classification in terms of well-known evaluation metrics.

1. Which abstract reports a primary research study? [1]
2. What gap/limitation in existing technology is abstract 2 trying to address? Write in one sentence only. [1]
3. As per abstract 1, what rationale/reason have the researchers given for their research problem? [1]
4. Write the research problem of abstract 2 in your own words. You can use the technical vocabulary, but not the academic vocabulary. [2]
5. As per abstract 2, what is the writers' verdict about the usefulness of their proposed solution? [1]
6. According to abstract 1, what broad themes were identified by the researchers while performing the review of literature? Name all of them. [2]
7. How is an adware different from a malware? Use the information in abstract 2 to answer this question. [2]

CLO #2: Compose reports for effective performance in professional life.

Q2.a: Write two problem statements/project definition using the following information. You are given information for two problem statements. Assume that you have completed both the projects. Be creative. Your research problem statement must solely focus on the project you have developed and the issue it solves. Thus, do not use more than 2 sentences to write the problem statement; however, you can provide elaboration wherever necessary. In addition, also write only 3 objectives for each problem statement. [10 marks]

a) Information for Problem Statement 1: Movie recommendation---watch history---user's preferences---Machine Learning algorithms---personalized movie recommendation system
b) Information for Problem Statement 2: Municipal Authority's Citizen Support System-----mobile app for citizens-----for complaints------management of complaints

Q2.b: Read the extracts given below and identify to which section they belong in a project reports. Choose from the options below. Write the extract number and the correct heading for it in the answer copy. [5 marks]

Recommendations---Stakeholders---Scope---Limitation

Extracts
1. regulatory authorities and policymakers may also be interested in the implications of predictive modeling on consumer privacy and data protection, highlighting the importance of ethical considerations in data science research and practice.
2. the research will investigate the effectiveness of various predictive modeling techniques, such as logistic regression, decision trees, and neural networks, in identifying potential churn indicators and predicting customer attrition.
3. the scalability of the routing algorithm was tested on moderately sized networks with up to 100 nodes, but its performance on larger-scale networks remains unexplored.
4. there is a need for research addressing the interpretability and explainability of deep learning models, as this remains a critical challenge in deploying these models in domains such as healthcare and finance where transparency and accountability are paramount.

5. Due to being coded using the commonly accessible technologies of PHP and MySQL our application is easily expandable to meet changing needs. Due to the modular implementation of the project , it would be easy to add new features with only minor changes to existing files.

CLO #3: Design, document, and develop a research project.

Q3: Given below is an extract taken from the "Literature Review" section of a research paper. Read it and answer the questions that follow. [10 marks]

Dust storms are one of the most dangerous natural hazards and rank only after extreme heat/cold and floods for causing the most deaths and injuries [1]. An increasing number of dust storm events have caused an increase in the reports of the loss of lives and property damages since 1995 (Figure 1). Death sand injuries are usually caused by car accidents, because of the reduced visibility during dust storm events [2]. A dust storm forecasting system would be able to alert the public of an approaching dust storm and can help people make safer plans before the potential event. That being said a dust storm forecasting system would be very helpful to enable the public to better prepare for such deadly events and to help reduce potential property damages [2, 3].

Dust storm events pose significant annual variability, for example dust storms usually occur during May and June and from September to November in southwest United States [3, 4]. Typically, most dust storms last from several hours to 1 day. Figure 2 shows the reported yearly frequency and percentage of the total time of dust storm in the United States [4]. It is estimated that the total time of dust storms was generally less than 90 hours and only taking less than 1% of 1 year, if assumed that each dust storm lasts an average of 2 hours. Therefore, a forecasting system for such events would require different computing and access requirements during different times of a year and even different hours within a day [5]. During dust storm events, massive concurrent user accesses to high-resolution dust storm forecasting are expected. The computing platform supporting such an emergency response system should be able to scale up enough computing resources to conduct relevant science, application, and education for the public to make smarter decision for saving more lives and reducing assets loss [5]. After the emergency response, the access to such information will be down and computing resources should be leveraged for other science, application, and education purposes.

Cloud computing is such a computing platform that its computing capabilities can be rapidly and elastically provisioned, in some cases automatically, to scale up and rapidly released to scale down. For different sizes of dust storm forecasting (sizes are determined by the spatial domain, resolution, and temporal scope), different amounts of computing resources are required [5, 6]. For example, 10 KM spatial resolution requires one CPU core, while 4 KM requires at least 16 CPU cores to complete the computing within 2 hours [Figure 3, 5]. Therefore, different amounts of computing resources should be allocated for different problem sizes [6]. This also causes disruptive computing requirements that can be best handled by elastic and on demand cloud computing platforms [7]. The emergency response systems, such as dust storm forecasting, require elastic large-scale computing pool [6, 7]. On the other hand, with the development of cloud computing technologies, for example virtualization, and the advancement of cloud infrastructure, cloud computing platforms are ready to support data and computing-intensive applications. For example, Amazon EC2 offers cluster instances with 10Gbps network connection [8]. Each cluster instance has minimum 23Gbytes memory, a clock speed of 2.93 GHz, and two quad-core processors. Such hardware and network configurations are far better than grid computing environments with heterogeneous computing resources and Wide Area Network (WAN) connection, and even better than most private homogenous high performance computing (HPC) cluster configurations. This makes cloud computing a new advantageous computing paradigm to resolve scientific problems traditionally requiring a special high-performance cluster [9]. The traditional cluster environment requires typically several months and usually hundreds of hours of labor to procure, install, and configure new infrastructure [9]. Traditional processes will involve multiple personnel with

different technical backgrounds to ensure the success of the configuration. Oftentimes, greater efforts and costs are required to maintain the infrastructure while the utilization of the infrastructure is only 10-15% of its full capacity [9]. Therefore, cloud computing would have a great impact on parallel scientific HPC applications in the near future by facilitating the deployment process while increasing the rate of utilization of computing resource [15].

1) Based on the above review of literature, which research problem can the researcher logically derive? [1 mark]
    a. This study aims to further advance existing grid computing environments used for the detection of dust storms.
    b. This study tests the feasibility of using cloud computing to support dust storm forecasting based on Amazon EC2 cloud computing platform. It focuses on costs and capacity assessment.
    c. This study aims to further advance existing grid computing environments used for the detection of dust storms by employing cloud computing. It tests the feasibility of such a system.
2) What line of argument has the writer followed to present his conceptualization/understanding of literature? [1 mark]
    a. Need for dust storm forecasting system---challenges associated with dust storm forecasting---strengths of existing methods used for forecasting---potential of cloud computing for forecasting
    b. Need for dust storm forecasting system---challenges associated with dust storm forecasting---complex computing requirements---potential of cloud computing for forecasting
    c. Need for dust storm forecasting system---challenges associated with dust storm forecasting---strengths of existing methods used for forecasting---potential of combining cloud computing with existing methods
3) Identify the statements as true or false based on the above review of literature. [5 marks]
    a. Total dust storm time is generally estimated to be 90 hours all over the world.
    b. Dust storm forecasting requires powerful computing resources.
    c. One of the strengths of cloud computing is its ability to address unexpected and changing computing requirements.
    d. Grid computing environments are a better option due to heterogeneous computing resources and Wide Area Network.
    e. Traditional processes deliver up to 20% of their full capacity.
4) In list format, write all the information borrowed by the writer from source [6]. [1 mark]
5) Name any two challenges associated with traditional approaches to forecasting dust storms? [1 mark]
6) Name any two reasons for using cloud computing to predict dust storms? [1 mark]

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